Sarvesh Baskar

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Education

Birla Institute of Technology and Science (BITS) Pilani — Goa, India
Dual Degree: Bachelor of Engineering in Computer Science & Master of Science in Physics — *Sep. 2020 – July 2025*

Experience

AI Research Assistant — Jul. 2025 – Present
University of Maryland, College Park — *Remote*

- Developed hypernetwork architectures for Frames2LoRA, predicting SmolVLM2 LoRA weights from intermediate video representations to avoid context-heavy visual token concatenation at query time.
- Implemented OpenQwen2VL textual embedding refinement experiments for VisAlign to mitigate object hallucination and improve visual grounding in large vision-language models.
- Ran controlled experiments, ablations, and Slurm-based training and evaluation workflows while tracking configurations, metrics, and results for reproducible multimodal reasoning research.

AI Consultant — Aug. 2025 – May 2026
Avyott — *Remote*

- Designed and deployed multimodal RAG chatbots using LlamaIndex, supporting grounded Q&A over enterprise text, image, and table documents for active clients at IPCA and Taj.
- Built Model Context Protocol servers in Python and Node.js to integrate chat workflows with ticketing systems and intent clarification, reducing task completion time by over 30%.

AI Research Assistant — Aug. 2024 – Jul. 2025
University of Maryland, Baltimore County — *Maryland, USA*

- Led experiments, analysis, and writing for a first-author NAACL 2025 SRW paper on resolving persona knowledge gaps in multi-turn LLM conversations through targeted clarification.
- Developed PlanForge, an Architect-Builder-Runner agent system that generates verified Python solver scripts from natural-language planning problems, reaching 100% success on NaturalPlan.
- Led AI modeling and evaluation for LLVM-IR code deobfuscation, reducing cyclomatic complexity by 85% on BCF-CFF samples with chain-of-thought prompting and graph analysis.

AI / Software Intern — May 2024 – Jul. 2024
Techisy — *Remote*

- Built a duplicate bill detection system using OCR, OpenAI embeddings, and CLIP to match visual and textual duplicates, achieving 95% identification accuracy.
- Developed a RAG-based candidate-to-job matching pipeline using LangChain, Ollama, and FAISS; deployed interactive Streamlit prototypes, reducing manual screening effort by 40%.

Selected Projects

Frames2LoRA | *SmolVLM2, LoRA, Hypernetworks, PyTorch, Python* — Mar. 2026 – May 2026

- Designed a perceiver hypernetwork that outputs LoRA adapters in a single forward pass, eliminating visual tokens from query context and enabling composition of independent video chunk adapters in rank space.

PlanForge | *LLM Agents, Planning, Code Generation, Verification* — May 2025 – Present

- Built an execution-grounded planning pipeline that converts natural-language constraints into solver programs, using iterative code-test-fix loops and deterministic validation for multi-constraint tasks.

Enterprise RAG and MCP Automation | *Langchain, MCP, Python* — Aug. 2025 – Nov 2025

- Designed RAG based chatbots for enterprise clients, enabling grounded Q&A over text, and table documents, and integrated MCP servers with ticketing workflows to reduce task completion time by over 30%.

Document Intelligence and Candidate Matching | *OCR, LangChain, FAISS, Streamlit* — May 2024 – Jul. 2024

- Built duplicate bill detection and candidate-to-job matching systems using OCR, OpenAI embeddings, CLIP, LangChain, and FAISS, achieving 95% duplicate identification accuracy and reducing manual screening effort by 40%.

Publications

Peer-Reviewed Publications

- Agrawal, Aakriti, Gouthaman KV, Rohith Aralikatti, Gauri Jagatap, Jiaxin Yuan, **Sarvesh Baskar**, Vijay Kamarshi, Andrea Fanelli, and Furong Huang. "Towards Mitigating Hallucinations in Large Vision-Language Models by Refining Textual Embeddings."
  ACL Findings, 2026.
- **Baskar, Sarvesh**, Manas Gaur, Srinivasan Parthasarathy, and Tanmay Tulsidas Verlekar. "(CPER) From Guessing to Asking: An Approach to Resolving Persona Knowledge Gap in LLMs during Multi-Turn Conversations."
  NAACL SRW, 2025.
- **Baskar, Sarvesh**, Muhammad R. Islam, Zikui Cai, Ankit Nakhawa, Anirudh Satheesh, Tom Goldstein, and Furong Huang. "The Low-Frequency Trap: Why Scaling Doesn't Solve Simple Temporal Counting."
  ICLR 2026 Workshop on I Can't Believe It's Not Better, 2026.
- Mohseni, Seyedreza, **Sarvesh Baskar**, Arman Hossain, Aritran Piplai, Edward Raff, and Manas Gaur. "Analyzing Chain of Thought (CoT) Approaches in Control Flow Code Deobfuscation Tasks."
  WoRMA, 2026.

Preprints and Under Review

- **Baskar, Sarvesh***, Zikui Cai*, Shayan Shabihi*, Anirudh Satheesh, Muhammad R. Islam, Udari Madhushani Sehwag, Tom Goldstein, and Furong Huang. "The Low-Frequency Trap: Video–Language Models Fail at Simple Event Bookkeeping."
  Under Review AAAI, 2027.
- **Baskar, Sarvesh**, Raviteja Bommireddy, Saksham Kumar Sharma, Harsha Kokel, Manas Gaur, Michael Katz, Shirin Sohrabi, and Kavitha Srinivas. "PlanForge: Synthesizing Planners from Natural Language with an LLM-Driven Architect–Builder–Runner Pipeline."
  Under Review AAAI, 2027.
- Suri, Manan, **Sarvesh Baskar**, and Dinesh Manocha. "Frames2LoRA: Parametric Video Internalization for Vision-Language Models."
  Under Review EMNLP, 2026.

Technical Skills

Languages: Python, C/C++, Bash, SQL, LaTeX, HTML/CSS
Frameworks & Libraries: PyTorch, Transformers, vLLM, LangChain, LlamaIndex, DSPy, RAG, LoRA, VLMs, OpenCV, TensorFlow, HuggingFace
Developer Tools & Systems: Git, Docker, Slurm, Redis, ChromaDB, FAISS, Ollama, MCP, Linux